

ENERGY CONSUMPTION FORECASTING AND OPTIMIZATION

Project Report submitted in partial fulfilment of the requirement for the award of the degree of Bachelor of Science in Computer Science with Artificial Intelligence and Data Science To Bharathiar University Coimbatore.



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DEPARTMENT OF COMPUTER SCIENCE

RATHINAM COLLEGE OF ARTS AND SCIENCE (AUTONOMOUS)

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Under section 2(f) & 12(B) and ISO 9001:2008 certified Institution)

(Affiliated to Bharathiar University)

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April – 2026

CERTIFICATE



This is to certify that the Project work entitled “**ENERGY CONSUMPTION FORECASTING AND OPTIMIZATION**” submitted to the Bharathiar University, in partial fulfilment of the requirements for the award of the Degree of **BACHELOR OF SCIENCE IN COMPUTER SCIENCE WITH ARTIFICIAL INTELLIGENCE AND DATA SCIENCE** is a record of original project work done by **T S PAMIKA (RCAS2023BAI028)** during the period 2023-2026 of her study in the Department of Computer Science, at Rathinam College of Arts and Science (Autonomous) under my supervision and guidance and the project has not formed the basis for the award of any degree other similar title to any candidate in any university.

Date:

Place: Coimbatore

Signature of the Guide

Signature of the Coordinator

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DECLARATION

I, **T S PAMIKA (RCAS2023BAI028)** hereby declare that this Project Report entitled “**ENERGY CONSUMPTION FORECASTING AND OPTIMIZATION**” , is the record of the original work done by me under the guidance of **Ms. M. YUVA BHARATHI, M.Sc.** Associate Professor, Rathinam College of Arts and Science, Coimbatore. To the best of my knowledge, this work has not formed the basis for the award of any degree or a similar award to any candidate in any University.

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DEPARTMENT OF COMPUTER SCIENCE

Project Completion Certificate

This is to certify that **T S PAMIKA (RCAS2023BAI028)**, Student of Rathinam College of Arts and Science (Autonomous), Coimbatore – 21 has successfully completed her project entitled "**ENERGY CONSUMPTION FORECASTING AND OPTIMIZATION**" during the academic year 2023 – 2026 under the supervision and guidance of **Ms. M. YUVA BHARATHI, M.Sc.** Assistant Professor, Department of Computer Science.

The conduct and character of the students during the period was good.

Signature of the Mentor

ACKNOWLEDGEMENT

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CONTENT

CHAPTER NO	CONTENT	PAGE NO
	ACKNOWLEDGEMENT	i
	LIST OF ABBREVIATION	iv
	LIST OF FIGURES	v
	ABSTRACT	vi
1	INTRODUCTION	
	1.1 About the Project	02
	1.2 Module Description	03
	1.3 Software & Technology Overview	04
	1.4 ER Diagram	04
2	SYSTEM ANALYSIS	
	2.1 Existing System	05
	2.2 Proposed System	05
	2.3 Feasibility Study	06
	2.4 System Specification	06
3	METHODOLOGY	07
4	SYSTEM SPECIFICATION	
	4.1 Hardware Specification	09
	4.2 Software Specification	09
5	SYSTEM ARCHITECTURE	
	5.1 System Architecture Diagram	12

6	SYSTEM TESTING	
	6.1 Data Flow Diagram (DFD)	14
7	SYSTEM IMPLEMENTATION	15
8	RESULT & DISCUSSION	
	8.1 Energy Consumption Dashboard	17
	8.2 Peak Usage Analysis Chart	17
	8.3 Cost Analysis Charts	18
	8.4 AI Insight	18
	8.5 ML Visualization Outcomes	19
9	CONCLUSION AND FUTURE ENHANCEMENT	22
10	APPENDIX	23
11	BIBLIOGRAPHY	25

LIST OF ABBREVIATIONS

SQL	Structured Query Language
BI	Business Intelligence
ER	Entity Relationship
HDD , SSD	Hard Disk Drive and Solid-State Drive
RAM	Random-Access Memory
GB	Giga Byte
OS	Operating System
ML	Machine Learning
EDA	Exploratory Data Analysis
DFD	Data Flow Diagram
RMSE , MAE	Root Mean Squared Error and Mean Absolute Error
AI	Artificial Intelligence
IoT	Internet of Things
LSTM	Long Short-Term Memory
INR	Indian Rupee
kWh	Kilowatt-Hour

LIST OF FIGURES

FIGURE NO	FIGURES	PAGE NO
Figure 1.4.1	Entity Relationship Diagram (ER Diagram)	4
Figure 5.1.1	System Architecture Diagram	12
Figure 6.1.1	Data Flow Diagram (DFD)	14
Figure 8.1.1	Energy Consumption Dashboard	17
Figure 8.2.1	Peak Usage Analysis Chart	17
Figure 8.3.1	Cost Analysis Charts	18
Figure 8.4.1	AI Insight	18
Figure 8.5.1	Actual vs Predicted Energy	19
Figure 8.5.2	Renewable Energy Recommendation Distribution	19
Figure 8.5.3	Correlation Analysis of Energy Usage and Cost Factors	20
Figure 8.5.4	Tariff vs Energy Usage	21
Figure 8.5.5	Energy Consumption by Consumption Level	21

ABSTRACT

Energy consumption has increased significantly due to population growth, urbanization, and technological advancements. Efficient energy management has become essential to reduce costs and environmental impact. This research presents a machine learning-based system for energy consumption forecasting and optimization using survey-based data.

Unlike traditional approaches that rely on secondary datasets, this study utilizes primary data collected through surveys to capture real-world energy usage behavior. The data includes factors such as user type, occupancy, appliance usage, renewable energy adoption, and consumption habits. The system uses SQL for structured data storage, Python for preprocessing and machine learning model development, and Power BI for visualization. Models such as Linear Regression and Random Forest are applied to predict future energy consumption. The results help identify peak demand, inefficient usage, and optimization opportunities.

The proposed system supports cost reduction, efficient resource utilization, and sustainable energy management, making it suitable for real-world applications.

The project uses “**Power BI**” as front end and “**Machine Learning**” and “**SQL**” as back end.

Keywords: *Energy Consumption Forecasting, Machine Learning, Energy Optimization, Data Analytics, Python, SQL, Power BI, Predictive Analysis, Time Series Data, Sustainability, Energy Management System, Data Visualization.*

CHAPTER 1

INTRODUCTION

Energy is one of the most fundamental resources that drives modern civilization, playing a crucial role in industrial development, economic growth, and daily human activities. With the rapid increase in population, urbanization, and technological advancements, global energy demand has increased significantly. This rising demand has created serious challenges such as resource depletion, increased electricity costs, and environmental issues including carbon emissions and climate change.

Traditional energy management systems are not capable of handling these challenges effectively. These systems rely heavily on static data analysis and lack the ability to process large-scale datasets or identify complex patterns in energy consumption. As a result, they fail to provide accurate forecasts or actionable insights for optimization.

To address these limitations, this project introduces an intelligent system that leverages machine learning and data analytics techniques to forecast energy consumption and optimize usage. The system is designed to analyze real-world survey-based data, which captures actual user behavior, making the analysis more practical and reliable.

The system integrates:

- **SQL** for structured data storage
- **Python** for data processing and machine learning
- **Power BI** for visualization

This combination enables accurate forecasting, better decision-making, and improved energy efficiency.

1.1 ABOUT THE PROJECT

The Energy Consumption Forecasting and Optimization system is a comprehensive data-driven solution designed to improve energy efficiency and reduce wastage. The project focuses on analyzing both historical data and behavioral factors to generate accurate predictions and meaningful insights.

Unlike conventional systems that rely on generalized datasets, this project uses **survey-based real-world data**, which provides a deeper understanding of how different factors influence energy consumption. These factors include user type, number of occupants, appliance usage, peak usage time, and renewable energy adoption.

The system performs multiple key functions, each contributing to improved energy management:

- **Energy Forecasting:**

Machine learning models are trained on historical data to predict future energy consumption. This helps in anticipating demand and avoiding unexpected energy shortages or overloads.

- **Behavioral Analysis:**

The system analyzes user behavior patterns to identify how lifestyle choices and appliance usage affect energy consumption. This enables more personalized and accurate insights.

- **Optimization Recommendations:**

Based on the analysis, the system provides suggestions such as reducing unnecessary usage, shifting consumption to off-peak hours, and adopting energy-efficient practices.

Additionally, the project supports decision-making by providing data-driven insights through interactive dashboards. These dashboards allow users to explore different scenarios and understand the impact of their energy usage patterns.

1.2 MODULE DESCRIPTION

The system is divided into multiple interconnected modules. The **Data Collection Module** gathers survey-based data such as user type, appliance usage, and energy consumption details. This ensures that the system captures real-world behavior.

The **Data Storage Module** uses SQL databases to store structured data. It ensures efficient querying and data integrity.

The **Data Preprocessing Module** prepares raw data for analysis by:

- Handling missing values
- Removing inconsistencies
- Transforming data into usable formats

This step is essential because clean data improves model accuracy.

The **Machine Learning Module** applies algorithms like Linear Regression and Random Forest to predict future energy consumption. These models identify patterns and relationships in the data.

The **Analysis Module** focuses on:

- Identifying peak usage times
- Detecting high energy-consuming users
- Understanding usage trends

The **Visualization Module** uses Power BI dashboards to present insights. These dashboards include charts, graphs, and filters for better understanding.

1.3 SOFTWARE & TECHNOLOGY OVERVIEW

The system uses a combination of modern technologies.

SQL is used for data storage and management. It supports large datasets and complex queries.

Python is used for preprocessing and machine learning. Libraries like Pandas and Scikit-learn help in data analysis and model building.

Power BI is used for visualization. It converts data into interactive dashboards that support decision-making.

This technology stack ensures:

- **Scalability**, to handle large datasets
- **Accuracy**, through machine learning
- **Usability**, via dashboards

1.4.1 ER DIAGRAM:

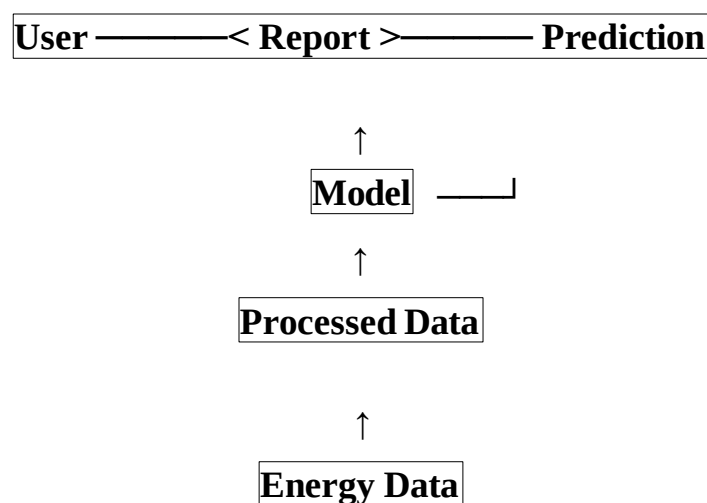


Figure 1.4.1 ER Diagram

CHAPTER 2

SYSTEM ANALYSIS

2.1 EXISTING SYSTEM

The existing energy management systems are mainly based on manual analysis and traditional statistical methods. These systems are not designed to handle large volumes of data or complex energy usage patterns. They rely on historical averages, which do not accurately represent real-time or dynamic changes in energy consumption.

One of the major limitations of these systems is the lack of predictive capability. They cannot forecast future energy demand, making it difficult for organizations to plan effectively. Additionally, these systems do not consider behavioral factors such as appliance usage and occupancy, which play a significant role in energy consumption.

As a result, organizations often face issues such as high energy costs, inefficient usage, and inability to identify peak demand periods. These limitations highlight the need for a more advanced and intelligent system.

2.2 PROPOSED SYSTEM

The proposed system addresses the limitations of the existing system by using machine learning and data analytics techniques. It provides a smart solution for forecasting energy consumption and optimizing usage.

The system uses predictive models to analyze historical data and forecast future energy demand. It also incorporates behavioral analysis by considering real-world user data, making the predictions more accurate.

In addition, the system provides interactive dashboards that allow users to visualize trends and insights. This makes it easier to understand complex data and make informed decisions.

The main advantages of the proposed system are:

- Improved accuracy in energy prediction
- Better understanding of consumption patterns
- Ability to identify and reduce energy wastage

This approach enables proactive energy management, where users can take action before problems occur.

2.3 FEASIBILITY STUDY

The feasibility of the system is evaluated in terms of technical, economic, and operational aspects. From a technical perspective, the system is feasible because it uses widely available tools such as Python, SQL, and Power BI. These technologies are easy to implement and support large-scale data processing.

From an economic perspective, the system is cost-effective. Most of the tools used are open-source or low-cost, and the system helps reduce energy expenses in the long run. From an operational perspective, the system is user-friendly. The dashboards are designed to be intuitive, allowing users to easily interpret the results and make decisions.

2.4 SYSTEM SPECIFICATION

2.4.1 Hardware Specification

- Processor: Intel i3 or above
- RAM: 4GB minimum
- Storage: 500GB HDD/SSD

2.4.2 Software Specification

- OS: Windows / Linux
- Tools: Python, SQL, Power BI
- Libraries:
 - 1) Pandas
 - 2) NumPy
 - 3) Scikit-learn

2.4.3 Input Design

Inputs include:

- User type
- Location
- Number of occupants
- Appliance usage
- Peak usage time
- Renewable energy usage

2.4.4 Output Design

Outputs include:

- Predicted energy consumption
- Consumption categories (Low/Medium/High)
- Cost analysis
- Dashboard insights

CHAPTER 3

METHODOLOGY

The methodology of the proposed system follows a structured data analytics workflow:

1. Collection of survey-based energy consumption data
2. Storage of data in an SQL database
3. Data preprocessing and feature extraction using Python
4. Machine learning model training and evaluation using

Linear Regression Model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

Random Forest Prediction

$$\hat{y} = \frac{1}{N} \sum_{i=1}^N T_i(x)$$

Mean Absolute Error

$$MAE = \frac{1}{n} \sum |y - \hat{y}|$$

5. Energy consumption forecasting and optimization analysis
6. Visualization using Power BI dashboards

CHAPTER 4

SYSTEM SPECIFICATION

4.1 HARDWARE SPECIFICATION

The hardware requirements for the Energy Consumption Forecasting and Optimization system are designed to ensure smooth execution of data processing, machine learning, and visualization tasks. Since the project involves handling datasets, running algorithms, and generating dashboards, a moderately capable system is sufficient.

A system with a minimum configuration of an Intel i3 processor, 4GB RAM, and sufficient storage can support basic operations. However, for better performance and faster processing, higher specifications such as 8GB RAM and a multi-core processor are recommended.

The hardware plays an important role in:

- Efficient execution of machine learning algorithms
- Faster data processing and analysis
- Smooth rendering of Power BI dashboards

Adequate storage is also necessary to maintain datasets, model outputs, and reports. Overall, the hardware requirements are affordable and easily accessible, making the system practical for real-world implementation.

In addition to the basic specifications, the system's performance can be further enhanced by incorporating a solid-state drive instead of a traditional hard disk, as it significantly improves data access speed and reduces loading time for datasets and applications. A stable internet connection is also beneficial for downloading datasets, accessing cloud-based tools, and updating software libraries. Furthermore, the use of a dedicated graphics processing unit is optional but can accelerate machine learning computations, especially when working with large datasets or more complex models. Proper hardware configuration ensures system reliability, minimizes processing delays, and supports scalability if the project is expanded in the future.

4.2 SOFTWARE SPECIFICATION

The software requirements include tools and platforms used for data storage, analysis, and visualization. The system is built using widely available and reliable technologies.

Python is the primary programming language used for data preprocessing and machine learning. It supports powerful libraries such as Pandas for data manipulation, NumPy for numerical operations, and Scikit-learn for implementing machine learning models.

SQL is used for storing and managing structured data. It allows efficient querying and supports large datasets.

Power BI is used for creating dashboards and visualizing insights. It helps in presenting complex data in an easy-to-understand format.

The software environment provides:

- Flexibility in handling different types of data
- Scalability for large datasets
- Integration between data processing and visualization

These tools together create a strong foundation for building an intelligent energy management system.

CHAPTER 5

SYSTEM ARCHITECTURE

The proposed system follows a layered architecture that clearly separates data storage, analysis, prediction, and visualization, making the system easy to understand, maintain, and scale. At the data storage layer, survey-based energy consumption data is stored in a structured SQL database. This ensures reliable data management, quick retrieval, and the ability to handle large volumes of energy data over time.

The data processing layer uses Python to clean and prepare the data for analysis. This includes handling missing values, formatting data, and extracting time-based features such as daily and monthly usage patterns. These steps help in understanding how energy consumption changes over time. In the prediction layer, machine learning models such as Linear Regression and Random Forest are trained using the processed data. These models learn past energy usage patterns and are used to predict future energy consumption, identify peak usage periods, and detect unusual or inefficient energy behavior.

Finally, the visualization layer presents the results through Power BI dashboards. These dashboards display historical trends, future forecasts, and optimization insights in a clear and interactive visual format, helping stakeholders easily understand the results and make informed energy management decisions.

This layered system design improves clarity, scalability, and flexibility, allowing the system to be easily extended or enhanced in the future.

5.1.1 SYSTEM ARCHITECTURE DIAGRAM:

Layered Architecture diagram:

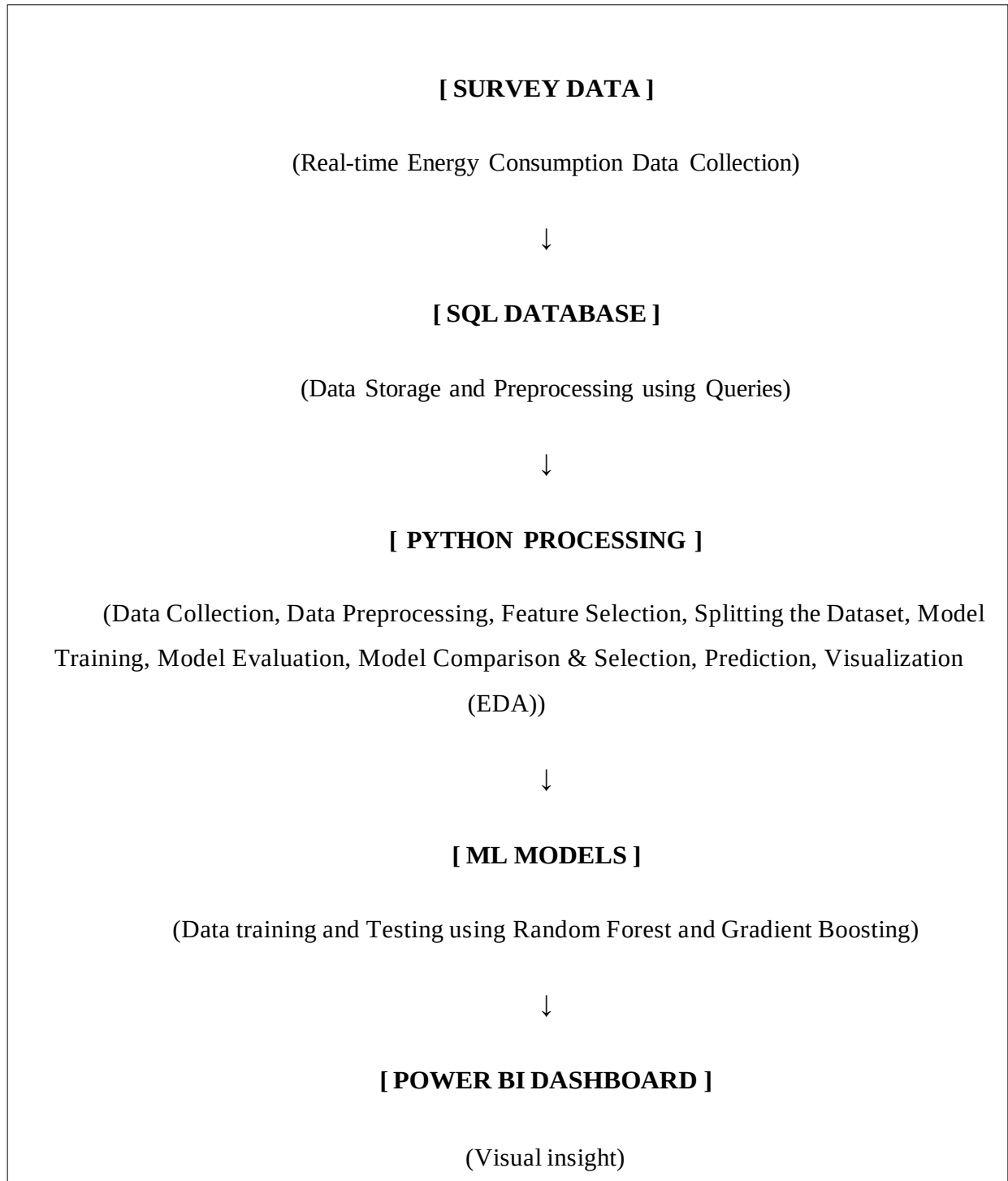


Figure 5.1.1 System Architecture

CHAPTER 6

SYSTEM TESTING

System testing is an important phase that ensures the system functions correctly and produces accurate results. The testing process verifies whether all modules are working as expected and whether the system meets the required objectives.

Different types of testing are performed to evaluate the system. Unit testing is used to check individual modules such as data preprocessing and machine learning. Integration testing ensures that all modules work together smoothly. System testing evaluates the overall performance of the system.

The testing process focuses on:

- Accuracy of predictions generated by machine learning models
- Proper functioning of data processing steps
- Correct display of results in dashboards

Performance metrics such as Mean Absolute Error are used to measure the accuracy of predictions. The results of testing indicate that the system performs reliably and produces meaningful insights.

Types:

- Unit Testing
- Integration Testing
- System Testing

Metrics:

- Accuracy
- Mean Absolute Error

6.1.1 DATA FLOW DIAGRAM (DFD):

Step No	Module/Process	Input Data	Process Description	Output
1	Data Collection	Raw energy data	Collect data from dataset/sensors	Raw Data
2	Data Preprocessing	Raw Data	Clean data, remove missing values	Clean Data
3	Feature Engineering	Clean Data	Extract useful features	Processed Data
4	Model Training	Processed Data	Train ML models (Regression, etc.)	Trained Model
5	Model Evaluation	Model + Test Data	Check accuracy (RMSE, MAE)	Performance Metrics
6	Prediction	New input data	Generate future energy usage predictions	Forecast Results
7	Visualization	Prediction Results	Create dashboards, charts	Reports

Figure 6.1.1 Data Flow Diagram (DFD)

CHAPTER 7

SYSTEM IMPLEMENTATION

System implementation involves the actual execution of the project, where all components are integrated to create a working system. This phase transforms the design into a functional application.

The implementation process begins with data collection through surveys. The collected data is then stored in an SQL database for structured management. Python is used to preprocess the data, including cleaning, transformation, and feature extraction.

After preprocessing, machine learning models are trained using the prepared dataset. These models analyze patterns and generate predictions for future energy consumption.

The final step involves visualization using Power BI, where dashboards are created to display insights such as consumption trends, peak usage, and cost analysis.

The implementation ensures:

- Smooth data flow between modules
- Accurate prediction and analysis
- User-friendly visualization of results

This structured approach makes the system efficient and easy to use.

CHAPTER 8

RESULTS & DISCUSSION

The system successfully analyzed energy consumption patterns and generated accurate predictions. The machine learning models demonstrated good performance in forecasting future energy usage.

One of the key findings was the identification of peak usage periods, particularly during evening hours. This insight is useful for managing energy demand and reducing load on the system.

Another important observation was the relationship between energy usage and cost. Users with higher consumption had significantly higher electricity bills, indicating the need for optimization.

The system also showed that users who adopted renewable energy had lower energy costs. This highlights the importance of sustainable energy practices.

Overall, the results confirm that machine learning can significantly improve energy management by providing accurate predictions and actionable insights.

The system successfully:

- Identifies peak usage patterns
- Predicts future consumption
- Detects inefficient

usage Findings:

- High usage occurs during peak hours
- Renewable energy reduces cost
- Strong relation between usage and billing

8.1.1 Energy Consumption Dashboard :

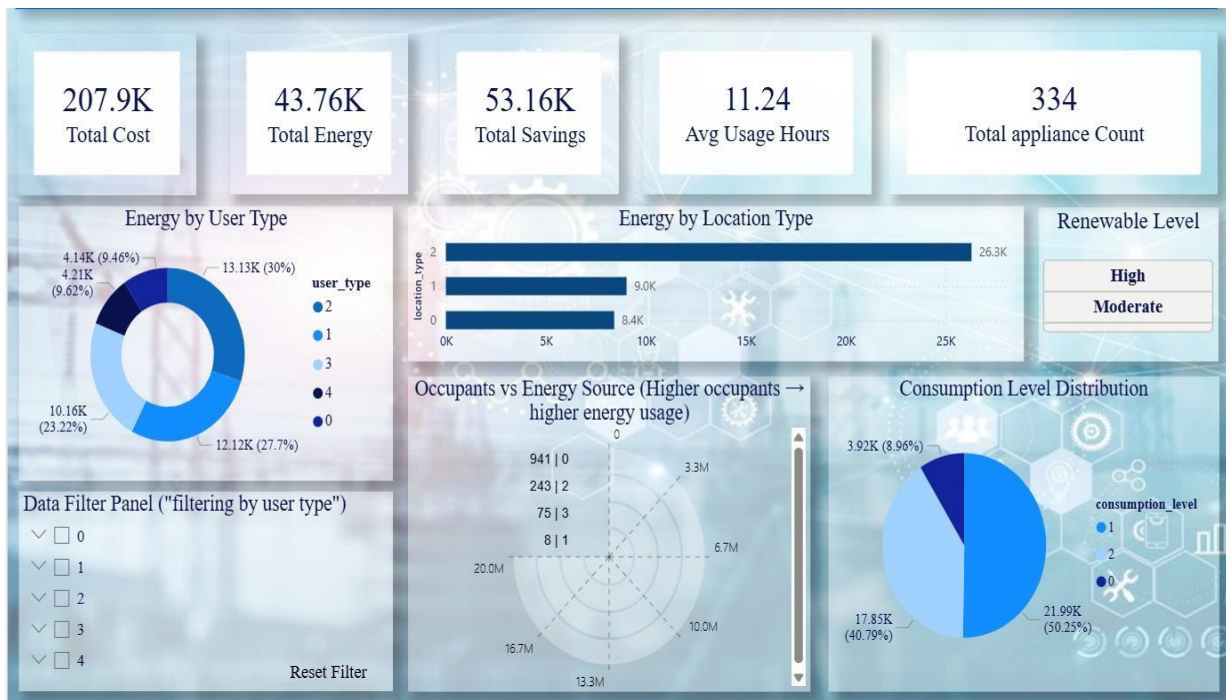


Figure 8.1.1 Energy Consumption Dashboard

8.2.1 Peak Usage Analysis Chart :

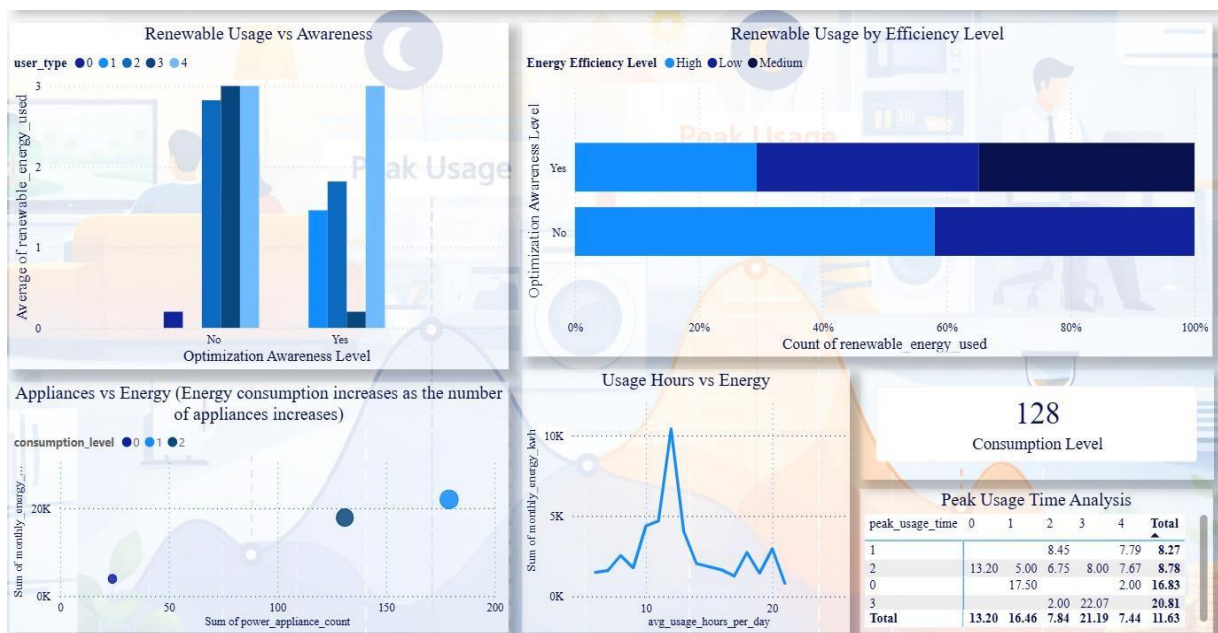


Figure 8.2.1 Peak Usage Analysis Chart

8.3.1 Cost Analysis Chart :



Figure 8.3.1 Cost Analysis Chart

8.4.1 AI Insight :

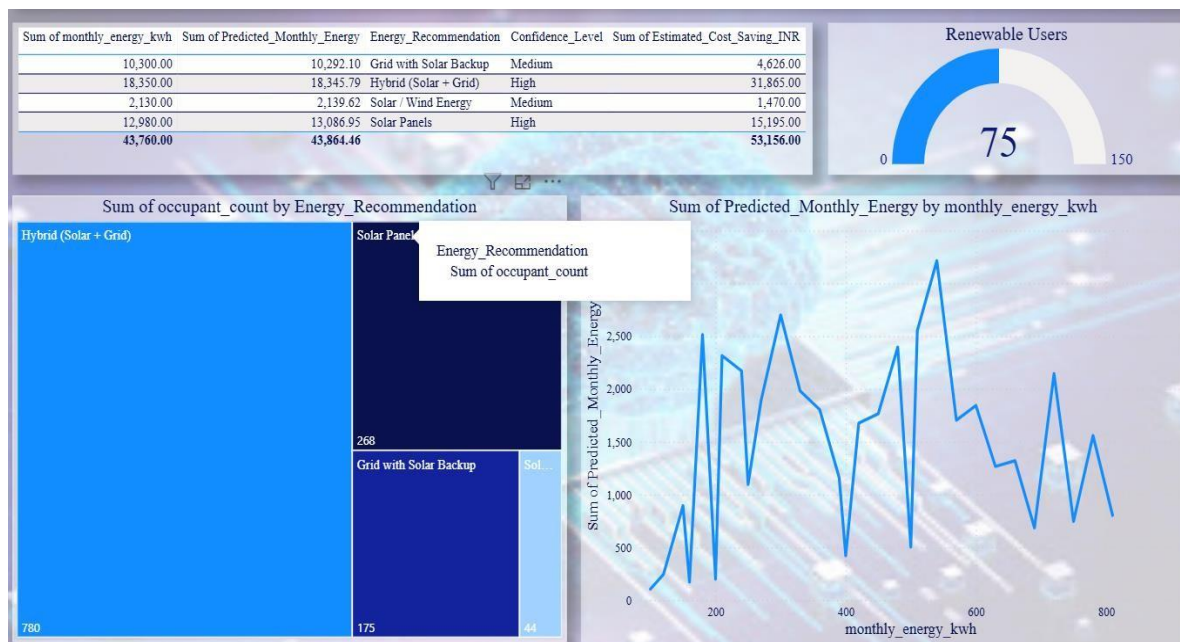


Figure 8.4.1 AI Insight

8.1 ML Visualization Outcomes :

8.1.1 Actual vs Predicted Energy :

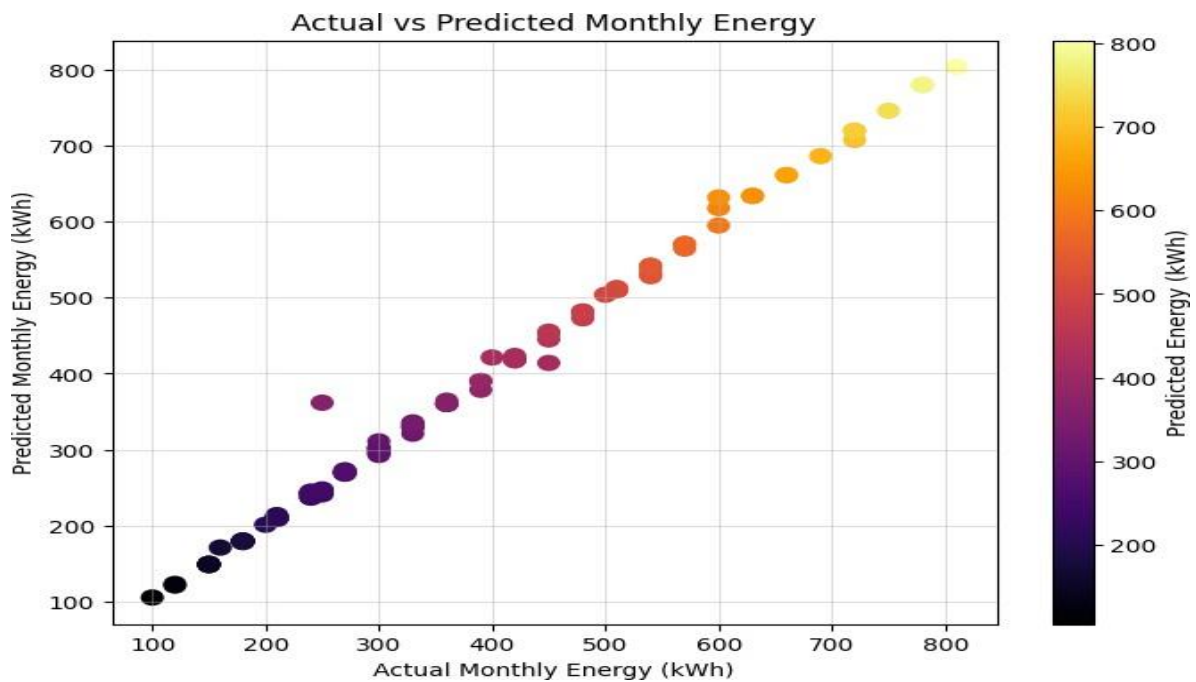


Figure 8.5.1 Actual vs Predicted Energy

8.1.2 Renewable Energy Recommendation Distribution :

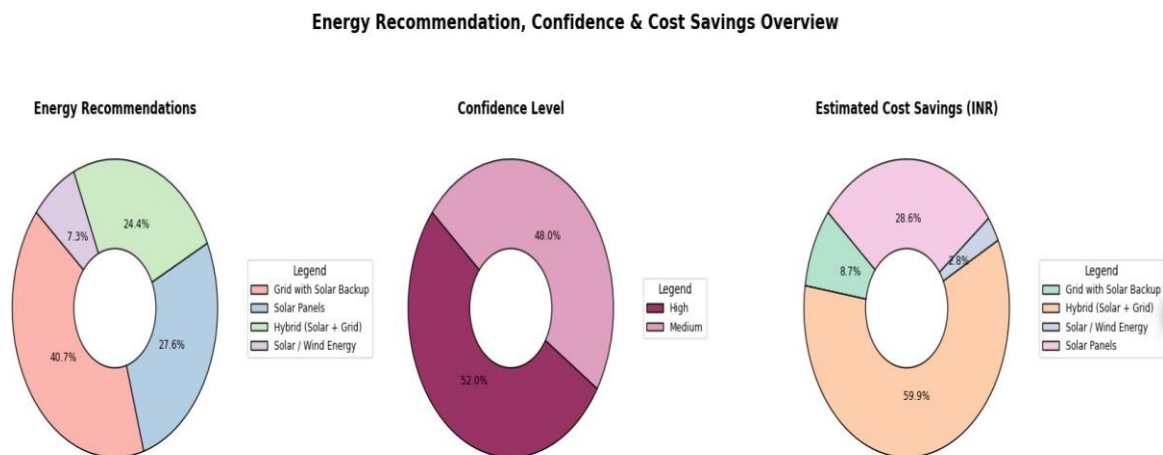


Figure 8.5.2 Renewable Energy Recommendation Distribution

8.1.3 Correlation Analysis of Energy Usage and Cost Factors :

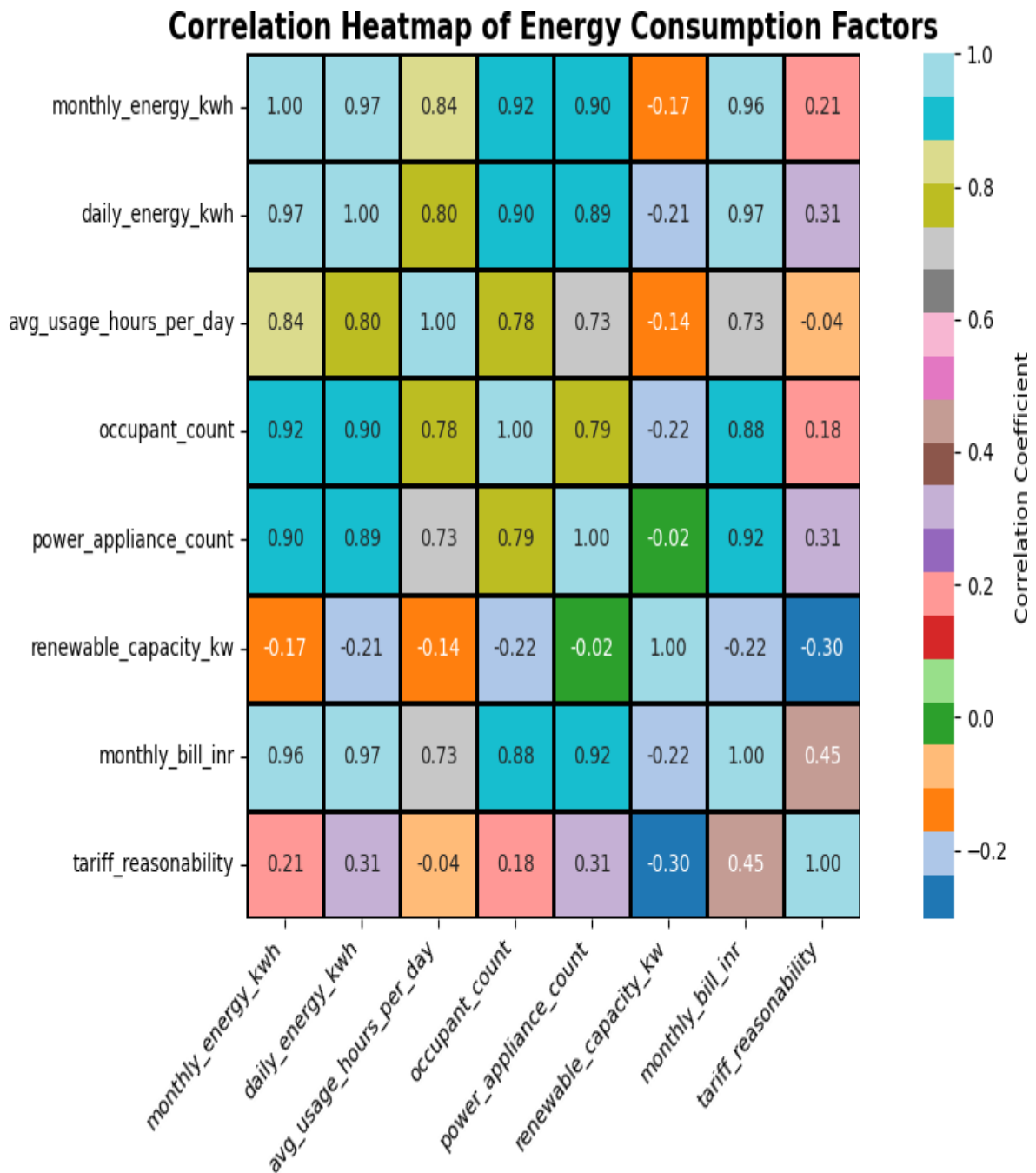


Figure 8.5.3 Correlation Analysis of Energy Usage and Cost Factors

8.1.4 Tariff vs Energy Usage :

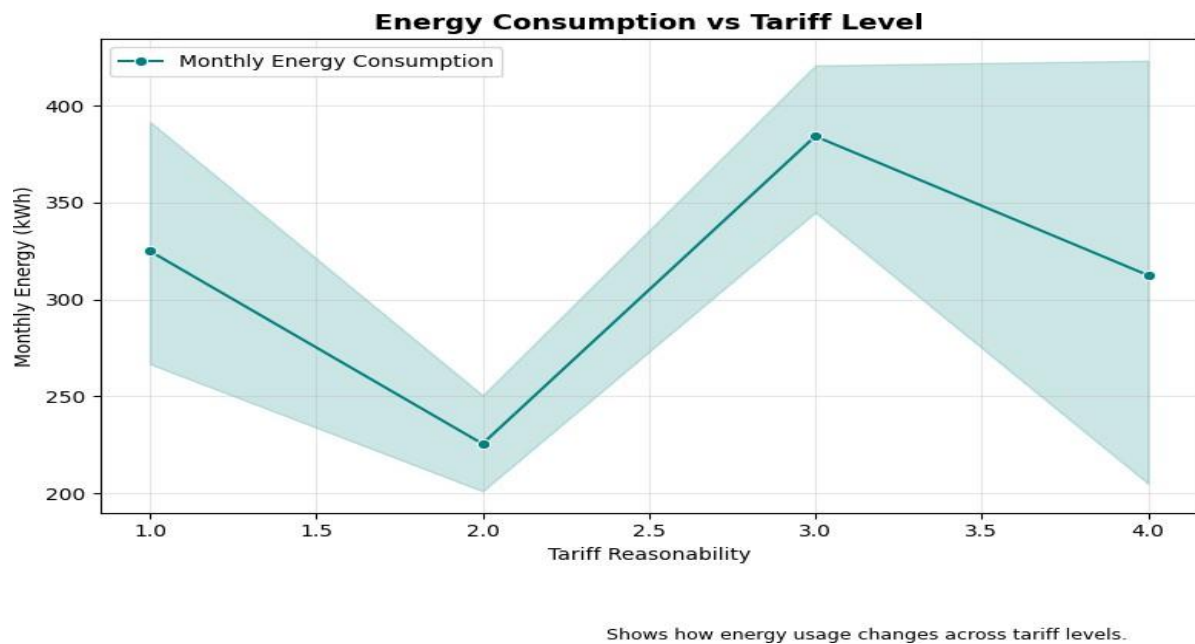


Figure 8.5.4 Tariff vs Energy Usage

8.1.5 Energy Consumption by Consumption Level :

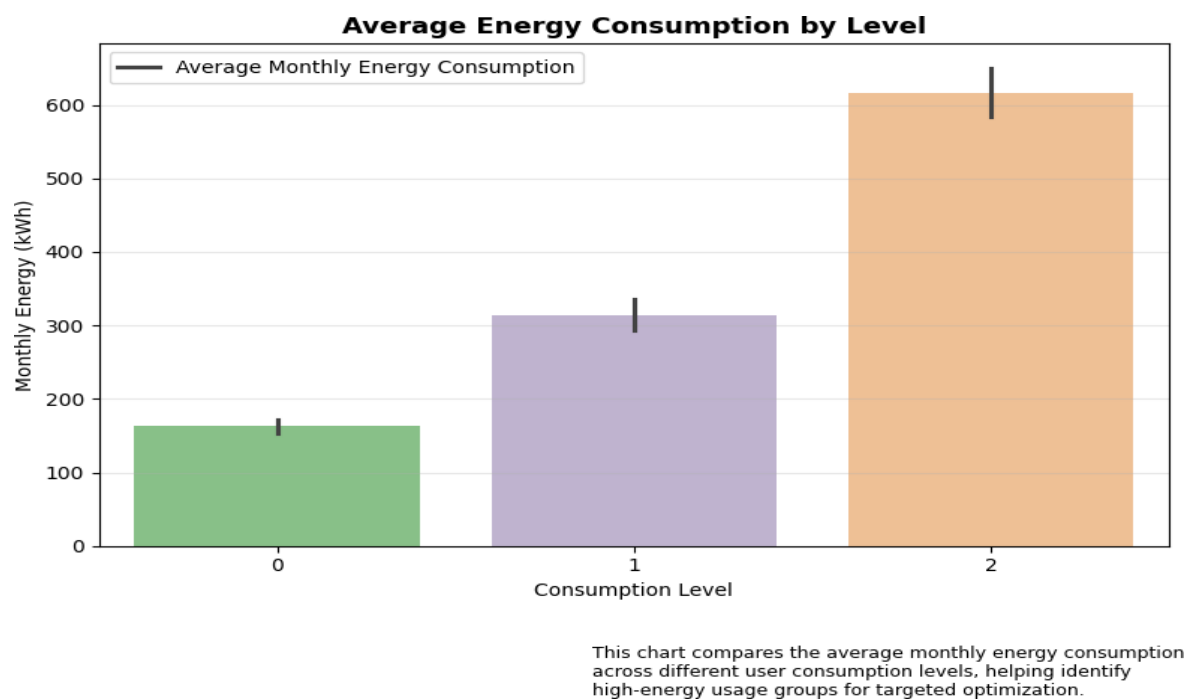


Figure 8.5.5 Energy Consumption by Consumption Level

CHAPTER 9

CONCLUSION AND FUTURE ENHANCEMENT

The Energy Consumption Forecasting and Optimization system successfully demonstrates how machine learning and data analytics can be applied to solve real-world energy management problems. The system effectively analyzes energy usage patterns, predicts future consumption, and provides optimization strategies.

By using survey-based data, the project ensures that the analysis reflects real-world conditions. The integration of SQL, Python, and Power BI creates a complete system that handles data storage, processing, and visualization.

The project highlights several key outcomes:

- Improved accuracy in energy forecasting
- Identification of inefficient energy usage
- Support for cost reduction and sustainability

Overall, the system provides a practical and scalable solution for energy management. It can be used in industries, institutions, and smart cities to improve efficiency and reduce energy wastage. The system can be further enhanced by integrating advanced technologies and expanding its capabilities. One major improvement is the use of IoT devices for real-time data collection. This would allow continuous monitoring of energy usage and immediate detection of abnormal patterns. Another enhancement is the use of advanced machine learning models such as LSTM, which are better suited for time-series data. These models can improve prediction accuracy by capturing complex patterns.

Additional improvements may include cloud integration for scalability, automated alert systems for user notifications, and mobile applications for better accessibility.

These enhancements will transform the system into a fully automated and intelligent energy management solution.

CHAPTER 10

APPENDIX

Appendix A: Sample Dataset Fields

- User Type
- Location Type
- Occupant Count
- Appliance Usage
- Peak Usage Time
- Renewable Energy Used
- Daily Energy (kWh)
- Monthly Energy (kWh)
- Monthly Bill (INR)

Appendix B: SQL Queries (Sample)

- Data cleaning and preprocessing queries
- Aggregation queries for consumption analysis
- Query for identifying high energy users
- Query for creating ML-ready dataset

Appendix C: Machine Learning Models Used

- Linear Regression Model
- Random Forest Model

Appendix D: Evaluation Metrics

- Mean Absolute Error (MAE)
- Prediction Accuracy

Appendix E: Power BI Dashboard Components

- Energy Consumption Trend Chart
- Peak Usage Analysis
- Renewable Energy Insights
- Cost Analysis Dashboard

Appendix F: Screenshots

- Dataset preview
- SQL query execution
- Model output results
- Power BI dashboard visuals

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